Kyle Rebello

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Education

McMaster University Hamilton, ON

Bachelor of Engineering in Computer Engineering *Expected Graduation: April 2029*

- **Awards**: Dean's Honour List (2025), Faculty of Engineering Award of Excellence (2024)
- **Relevant Coursework**: Digital Systems Design, Digital Logic Design, Algorithm Design and Analysis, Electronic Circuits and Devices II, Linear Algebra, Calculus III, Advanced Probability and Random Processes, Signals and Systems

Work Experience

ACTO May 2026 – Present

AI Engineer Intern *Toronto, ON*

- Cut median quote-build time **4×** (from **∼120 to 30 min**) for **10+ sales reps**, by replacing a legacy Excel pricing calculator with a natural-language LLM sales agent
- Ensured every quoted price is computed deterministically, by architecting a hybrid system in which a deterministic pricing engine calculates all figures and the LLM only narrates them, confining hallucination and prompt-injection risk to explanatory text
- Delivered citation-backed answers traceable to source documents, by building a retrieval-augmented generation (RAG) pipeline over an Amazon OpenSearch Serverless vector index
- Shipped the full-stack B2B platform hosting the agent, including SSO, three-tier role-based access control, PDF order-form generation, and token-secured proposal pages with engagement analytics

Weever Apr. 2026 – Present

Machine Learning Engineer (Part-time Contract) *Hamilton, ON*

- Enabled **zero-configuration** onboarding of any customer's operational logs, by building a schema-agnostic Python pipeline (pandas, SciPy) that infers column types from content rather than names, with seeded, reproducible scoring
- Segmented workforces into behavioural archetypes, by engineering **45 features** (submission cadence, evidence quality, measurement consistency) and selecting between k-means and Gaussian mixture models using silhouette scores in scikit-learn
- Gave managers a ranked watchlist of workers showing engagement drift, by building a composite risk score that compares each worker's output, documentation quality, and work patterns against their own baseline and their team's trend
- Let managers query results in plain English, by shipping the analytics layer as a self-contained local app with an MCP server for LLM-based natural-language querying

Projects

INT8 FPGA Neural Network Accelerator | *Verilog, PyTorch, Python, Quartus* Oct. 2026 – Present

- Built a hardware neural network accelerator that classifies handwritten digits in **128.7 μs** (**6,437** cycles at 50 MHz) at **97.46%** accuracy, by designing a 16-unit parallel multiply-accumulate (MAC) engine in Verilog for a Cyclone IV FPGA (DE2-115)
- Cut model memory **4×** with only **0.01%** accuracy loss, by quantizing a PyTorch-trained network from 32-bit floats to 8-bit integers using hardware-friendly power-of-2 rescaling
- Achieved a **100%** bit-exact match with a Python golden model on all **1,390** checked values (every hidden value, score, and digit), by building self-checking Verilog testbenches in ModelSim and a script that verifies the weight-memory layout
- Built a stand-alone board demo that shows each of **10** test digits on a VGA monitor and the network's guess on the 7-segment display, matching the golden model on **all 10** (including its one mistake), by designing a 640×480 VGA controller, on-chip image memory, and button-driven controller in Verilog

GPU Monte Carlo Circuit Simulator | *C++, CUDA, OpenMP, LTspice, Python* Oct. 2026

- Predicted the gain spread of a BJT amplifier across **1,000,000** random part combinations in **2.0 s** on a Jetson Nano GPU, by writing a SPICE-style C++/CUDA circuit solver that runs Newton's method for each random circuit on its own GPU thread
- Matched LTspice within **0.005%** on bias voltages, gain, and cutoff frequency, and LTspice's Monte Carlo averages within **0.2%**, by validating the simulator against it with an automated comparison script
- Sped up the GPU simulator **20.8×** (to **505,000** builds/s, **3.2×** faster than all 4 CPU cores), by cutting repeated work, folding the transistor's internal resistors into a smaller 5×5 system, and switching to 32-bit math with under **0.002%** error

Technical Skills

Languages: Python, C/C++, Verilog/SystemVerilog, SQL (PostgreSQL, SQLite), MATLAB
Machine Learning: PyTorch, scikit-learn, pandas, NumPy, SciPy, LLMs, RAG, vector databases, INT8 quantization
Cloud & Dev Tools: AWS (Bedrock, OpenSearch Serverless, ECS, ECR, RDS, S3), Docker, Linux, Git, CI/CD
GPU & Parallel Computing: CUDA, OpenMP, mixed-precision (FP32/FP64) optimization, performance benchmarking
Hardware: FPGA design (Cyclone IV), Quartus, ModelSim, NVIDIA Jetson Nano
Analog & Digital Electronics: Operational Amplifiers, Multistage Amplifier, Oscillators, Field-effect and Bipolar Transistors, CMOS circuits, Circuit Design & Analysis (Analog Discovery 3, PSpice, LTspice)

Extracurriculars

McMaster Dragon Boat Competitive Rowing Team, Men's Intramural Soccer